School of Engineering

Unit Outline

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TNE30009

Network Security and Resilience

Semester September, 2026

Please read this Unit Outline carefully. It includes:

PART A Unit summary

PART B Your Unit in more detail

PART C Further information



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PART A: Unit Summary

Unit Code(s)		TNE30009
Unit Title		Network Security and Resilience
Duration		12 weeks
Total Contact Hours		60
Requisites:		
	Pre-requisites	TNE10006 Networks and Switching OR COS20012 Communications and Security OR equivalent introduction to IP networking
	Co-requisites	
	Concurrent pre-requisites	
	Anti-requisites	
	Assumed knowledge	Basic understanding of IP networking
Credit Points		12.5
Campus/Location		Swinburne Vietnam
Mode of Delivery		On campus
Assessment Summary		Portfolio 80%, Online quizzes 20%

Aims

This unit of study aims to introduce you to the issues of security in modern networks.

Unit Learning Outcomes

Students who successfully complete this unit can:

1. Describe the threats to network security and their defences (K3)
2. Formulate an organisational security policy (S1, S2, A2)
3. Plan the implementation of a security using technologies such as firewalls, VPNs, Intrusion Detection Systems, and authentication systems (S1, S2, A2)
4. Conduct encryption of simple messages using public key and secret key techniques (K3)
5. Describe techniques used in dealing with Denial of Service Attacks (K3)
6. Describe the operation of the major security protocols (K3)
7. Describe the security issues relating to wireless technologies (K3)

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Swinburne Engineering Competencies for this Unit of Study

Swinburne Engineering Competencies for this Unit of Study

This Unit of Study will contribute to you attaining the following Swinburne Engineering Competencies:

K1	**Basic Science:** Proficiently applies concepts, theories and techniques of the relevant natural and physical sciences to the solution of complex engineering problems.
K2	**Maths and IT as Tools:** Proficiently uses relevant mathematics and computer and information science concepts as tools in complex engineering activities.
K3	**Discipline Specific:** Proficiently applies advanced technical knowledge of the specific discipline within that context.
K4	**Emerging Disciplinary Trends:** Proficiently applies research principles and methods on current or emerging complex problems of the specific discipline.
S1	**Engineering Methods:** Applies engineering methods in practical applications and complex engineering problems.
S2	**Problem Solving:** Systematically uses current or emerging knowledge and research methods to undertake independent research in solving complex engineering problems and as preparation for research higher degrees.
S3	**Design:** Systematically uses engineering methods in designing solutions to complex engineering problems.

Graduate Attributes

The Swinburne Graduate Attributes describe the capability of our graduates to use knowledge, skills and behaviours to contribute to society meaningfully and positively. They include professional, self-directed learning and future-ready skills.

Graduate attributes may be practised in the unit but are not formally taught or assessed as part of the unit content

Content

Threats to network security
Security policy
Firewalls
Encryption
Virtual Private Networks
Intrusion Detection Systems
Authentication Systems
Wireless Security

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PART B: Your Unit in more detail

Unit Improvements

Because network security is a constantly changing area, this unit is updated every year. Student feedback is also considered during the update. Changes this year include:

- Discussion of threat environment updated to include recent attacks and attack types
- Additional material on security protocols
- Updated material on cryptography
- Internet of Things

Unit Teaching Staff

Name	Role	Location	Email / Teams	Consultation Times
Dr. Pham Van Dai	Lead Coordinator	Ha Noi	dvpham@swin.edu.au	By email appointment
Dr. Sam Nguyen	Unit Coordinator	HCM City	sxnguyen@swin.edu.au	By email appointment

Learning and Teaching Structure

Category	Activity	Total Hours	Hours per Week	Teaching Period Weeks
In person	Lectures	36 hours	3 hours	Weeks 1 to 12
In person	Class	11 hours	1 hour	Weeks 2 to 12
In person	Lab	12 hours	1 hour (average)	Every second week from week 1
Live Online or in person	Consultation	12 hour	1 hour	Weeks 1 to 12

Week by Week Schedule

Week	Teaching and Learning Activity	Student Task or Assessment
1	Lecture: Overview, networks and TCP review	
2	Lecture and Tutorial: Threats Lab 1: Tunnelling	Lab assessed via a short quiz at end of session
3	Lecture and Tutorial: Threats	
4	Lecture and Tutorial: Security Policy, Authentication Lab 2: Firewalls	Lab assessed via a short quiz at end of session
5	Lecture and Tutorial: Firewalls	

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6	Lecture and Tutorial: VPNs, Intrusion Detection and Prevention Lab 3: VPNs	Lab assessed via a short quiz at end of session and report due in week 8
7	Lecture and Tutorial: Denial of Service attacks and defences	Mid semester test
8	Lecture and Tutorial: Cryptographic algorithms Lab 4: Authentication	Lab assessed via a short quiz at end of session Lab 3 report due
9	Lecture: Cryptographic algorithms	
10	Lecture and Tutorial: Security protocols Lab 5: Cryptography	Lab assessed via a short quiz at end of session and report due in week 10
11	Lecture and Tutorial: Security protocols, Wireless Security including IoT, Cloud Security	
12	Lecture and Tutorial: Wireless Security, Revision Lab 6: Cryptography	Lab 5 report due Lab assessed via a short report due in that week End of semester test worth 10%

Assessment

a) Assessment overview

To pass this unit, you must:

- Achieve an overall mark for the unit of 50% or more, and
- Satisfactorily complete all portfolio assessment items. This includes the research report and the case study. You must submit both to pass.

Students who do not complete all portfolio items will receive a maximum of 45% as the total mark for the unit.

Tasks and Details	Individual or Group	Weight	Mapped Unit Learning Outcomes	Mapped Graduate Attributes	Assessment Due Date
Research report of up to 2500 words	Individual	HD	1,2,3,4,5,6,7		During the exam period
Case study up to 2500 words	Individual	HD	1,2,3		Last day of semester
Short laboratory quiz upon completion (6)	Individual	P	1,3,4,6		End of each lab session
Laboratory reports (2)	Individual	P	1,3,4,6		End of week two weeks after lab session
Optional mid semester test (in-class)	Individual	10%	1,2,3,5		Week 7

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Optional end of semester test (in-class)	Individual	10%	1,2,3,5		Week 12

Assessment Requirements	Details
b) Use of generative AI (genAI) in this unit	The valid use of genAI in this unit is as follows: • genAI may be used for brainstorming, creating structures, and generating ideas for improving work. Any use of genAI must be acknowledged, with prompts and outputs included in an appendix.
c) Hurdle requirements	All assessments other than the online quizzes must be passed.
d) Final assessment period	You are expected to be available for the entire final assessment period including any Special Exam period.
e) Submission requirements	Assignments and other assessments are submitted online through the Canvas assessment submission system which integrates with the Turnitin . Please ensure you keep a copy of all assessments that are submitted.
f) Extensions and late submissions	Late Submissions - Unless an extension has been approved, late submissions will result in a penalty. You will be penalised 5% of total assessment value for each working day the task is late, up to a maximum of 7 working days. After 7 working days, a zero result will be recorded.
g) Referencing	To avoid breaching academic integrity, you are required to provide references whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as ChatGPT). Further details regarding academic integrity are available in Section C of this document. Referencing conventions required for this unit is IEEE. Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/
h) Groupwork guidelines	There is no groupwork for this unit.

Required Textbook(s)

There are no prescribed text books for this unit.

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Recommended Reading Materials

Swinburne Library has a large collection of resources. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.
The following are good references:

- Schneier, “Applied Cryptography” Second Edition 1996
- Trappe, Washington, “Introduction to Cryptography with Coding Theory”, 2006
- Anderson, “Security Engineering”, 2009
- Mell, “What’s Special about Cloud Security?”, IEEE IT Professional, July/August 2012
- Kaufman, “Can Public-Cloud Security Meet Its Unique Challenges?”, IEEE Security and Privacy, July/August 2010
- Granjal, Monteiro and Sa Silva, “Security for the Internet of Things: A Survey of Existing Protocols and Open Research Issues”, IEEE Communications Surveys and Tutorials, 17(3) 2015
- Tiburski, Amaral, deMatos, Azevedo, Hessel, “The Role of Lightweight Approaches Towards the Standardization of a Security Architecture for IoT Middleware Systems”, IEEE Communications, Dec 2016

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PART C: FURTHER INFORMATION



For further information on any of these topics, refer to Swinburne's Student webpage http://www.swinburne.edu.au/student/

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The Student Charter describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the Student Academic Misconduct Regulations, Student General Misconduct Regulations and the People, Culture and Integrity Policy. Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response.

Canvas

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the Student login webpage or https://swinburne.instructure.com/ Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence;

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contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Convenor.

Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarism, collusion, contract cheating, unauthorised file sharing, falsification, fabrication, manipulation or misrepresentation of information, reuse of previous work and non-compliance with instructions in an invigilated or non-invigilated assessment item are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include, but are not limited to:

- submitting work as your own for assessment that has been fully or partially completed by a third party, either paid or unpaid
- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Convenor
- using another person's work or ideas as though it is your own work, without appropriate attribution
- working closely with another student or group of students (either past or current), to submit for assessment, some or all of the other student or students' work as your own work
- sharing without permission of the Unit Convenor, Swinburne resources or other material related to assessment to an entity or document repository site
- creating, intentionally modifying or inventing information that is intended to be submitted as part of an assessment item
- using the whole or part of a computer program written by another person as your own without appropriate acknowledgement
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another person without acknowledgment
- enabling others to cheat, including letting another student copy your work or by giving access to a draft or completed assignment
- letting someone or something else impersonate you, or you impersonate someone else in an invigilated or non-invigilated assessment item
- accessing, obtaining and/or providing to others unauthorised materials relating to an invigilated or non-invigilated assessment item.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne. For further details, see https://www.swinburne.edu.au/student-login/academic-integrity/

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully. Your Unit Convenor or studentHQ can provide information about the study support and other services available for Swinburne students. For further information, see the Current students web page.

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Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration are submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. See https://www.swinburne.edu.au/life-at-swinburne/student-support-services/special-consideration-assistance/

Note: Submitting fraudulent (fake or altered) medical certificates is considered misconduct and can lead to serious penalties from Swinburne. In addition, your doctor may report fraudulent medical certificates as a prosecutable offence under the Victorian Crimes Act.

AccessAbility Services

If you are a student with a disability, medical or mental health condition or you have significant carer responsibilities, you may require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an Education Access Plan (EAP) that includes the services and reasonable adjustments that you need.

It is recommended that you register with AccessAbility Services when you first commence your course but you can contact the service at any time during your studies to find out about reasonable adjustments. Contact Accessibility Services to discuss further.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Convenor to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution. If you are dissatisfied with the outcome of the local resolution, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Convenor. If your concerns are not resolved or you would prefer not to deal with your Unit Convenor, then you can complete a feedback form. See https://www.swinburne.edu.au/corporate/feedback/

Advocacy

Should you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Academic Student Support Officer at Swinburne Student Life.

For an appointment, please call 19006412 or lodge a query via Student Portal at https://portal.swin.edu.vn

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